

Nail Disease Prediction using a Deep Learning Integrated Framework

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Abstract— Deformities that may develop in the nail unit are referred to as "nail diseases". Even though the nail unit is regarded as a skin accessory, these diseases have their own distinct set of signs, causes, and outcomes that may or may not be connected to other illnesses. It can be challenging to identify and recognize certain nail disorders, hence this research suggests a novel deep learning framework that can identify and categorize these diseases from images. The framework is specifically made to identify nine different types of nail diseases based on nail colour, including black, blue, grey, purple, red, white, and yellow coloured nails with beau's lines, hyperpigmentation, onychomycosis, subungual hematoma, yellow nail syndrome, psoriasis, koilonychia, paronychia, pincer nails, leukonychia, and onychorrhexis. In order to do this, feature extraction is carried out using a mix of convolutional neural networks (CNNs). The researchers created their own dataset to assess the effectiveness of their suggested framework because there isn't a comprehensive dataset readily accessible for this assignment. The outcomes were contrasted with those of other cutting-edge algorithms, including RF, SVM, ANN, and KNN, which are renowned for excelling in feature extraction. With an accuracy of 87.33%, the suggested framework distinguished between varieties of nail conditions with comparable performance.

Keywords— *Nail Image Processing, Deep Learning, Nail Diseases, Deep Convolutional Neural Networks, Feature Extraction.*

I. INTRODUCTION

Keratin, a hard protein acts as a protective coating for the tissues of the finger nails and toes. The condition of the nail unit might provide important information about a person's general health. The existence of different diseases, such as psoriatic arthritis, diabetes, anaemia, fungal infections, melanoma, hepatic disorders, and malnutrition, may be indicated by certain forms of nail discoloration, hyperpigmentation, changes in nail texture, and the presentation of particular conditions.

People can develop nail infections at any age [1], and onychomycosis, a painful fungal infection, affects around 15% of the population. The majority of nail conditions are contagious, and some, such subungual melanoma, can be fatal. This specific illness is challenging to identify in the early stages due to its two-year incubation period and 43% 10-year survival rate. There was not much study done on the identification of nail anomalies before to 2003 [2].

However, there has been an increase in interest in the study of nail disease detection recently, with many researchers working in this field. The nail unit has been used to diagnose illnesses, forecast and detect psoriatic arthritis, and analyse texture and colour [3, 4]. Additionally, studies

on early disease detection and associated subjects have been done.

Researchers' focus on the medical science subfield of dermatology is growing. Interest in a number of areas has recently increased, including image matching algorithms [5], classification of dermatological pictures, automated melanoma screening, detection of skin cancer and the identification of dermatological illnesses. Deep learning, a method that uses multiple-layer artificial neural networks, has become a hot issue, drawing many academics to investigate its possibilities. Skin cancer was classified by utilizing deep neural network by Esteva et al. [6], demonstrating the significance of this approach.

Deep learning approaches have shown noticeably better results than traditional models like Random Forest (RF), K Nearest Neighborhood (KNN) and Support Vector Machine (SVM). Deep learning techniques that use neural networks have shown to be capable of accurately approximating any function [7]. Additionally, studies have shown that adding more layers to the network improves performance more effectively than other changes [8].

Deep learning makes use of computer models with several levels of abstraction in the processing. These techniques have outperformed conventional methods in the area of visual object recognition and detection [9]. Deep learning uses the back-propagation process, which tells a computer to change the internal parameters used to compute the representation in each layer from the previous layer, to recognize complex patterns in large data sets. Deep convolutional networks have enabled considerable improvements in the processing of images and videos using these methods [10].

To perform various tasks, many deep learning models have been constructed. Neuroscience-based algorithms, sometimes referred to as neural networks, were created in the field of visual computing to mimic the operation and activity of the human visual cortex. These simulations imitate the brain's neurons, which conduct various levels of abstraction and combine the characteristics of earlier layers with those of the present layer. These models include convolutional neural networks (CNNs), which include numerous deep layers with various degrees of abstraction [11].

Because they can be trained on large datasets using powerful GPUs, CNN have gained popularity in object classification and feature learning [12]. This is because they perform better and are more accurate.

Identification of nail diseases is a relatively untapped subject. There has been minimal attention paid to diagnosing nail disorders, even though the nail unit has been inspected

and studied for a variety of reasons, including the identification of certain diseases like melanoma and psoriatic arthritis. Onychomycosis, Beau's lines, psoriasis, yellow nail syndrome, hyperpigmentation, koilonychia, paronychia, subungual hematoma, pincer nails, leukonychia, and onychorrhexis are among the nail illnesses we diagnose and categories using CNN.

There are a number of difficulties that must be overcome in the context of the current issue. For starters, there aren't many high-quality datasets available for the task. In addition, individual differences in nail form and texture necessitate very accurate and effective feature identification. We have created a cutting-edge deep learning framework that makes use of an ensemble of CNN to overcome these challenges.

Paper is structured in following manner: Introduction of topic is covered in Section 1, there after Section 2 represents the review of literature. Section 3 discusses the data collection process, while Section 4 details the methodology used in the study, including the classifiers and proposed approach. The results of the study and conclusions are presented in Section 5 and Section respectively.

II. REVIEW OF LITERATURE

In recent years, a lot of researchers have become interested in using computer vision methods for illness diagnosis, such as feature extraction and classification. The processing of human nail pictures, which may be used to recognize certain types of disorders, is a new subject in this area.

The human nail unit was used by Trupti et al. [13] to construct an early-stage illness recognition system utilizing image processing methods. Their accuracy was 65% using nail pigment as the distinguishing characteristic. Nail illnesses were one of the categories in the deep learning method for classifying global skin diseases that Liao and Haofu [14] presented.

For nail illnesses, there were, however, only a few pictures per class. Images produced from nail capillaroscopy were improved by Niraj and Gerald [15] using computer-aided nail fold capillaroscopy image analysis. By examining the health of the blood capillaries in the Nail fold, this technique can be implemented to identify conditions like Raynaud's syndrome and scleroderma. [16] uses a colour histogram technique to analyze psoriasis, extracting features using the HSV colour space, and achieving an average accurateness of 92%.

Our research uses a dataset of 18025 photos that we have gathered in an effort to create a deep learning architecture specifically designed for nail illnesses.

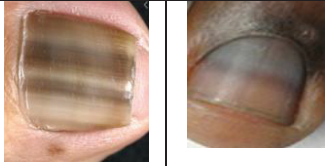
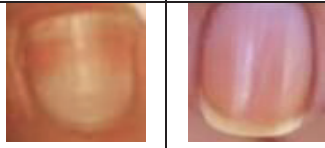
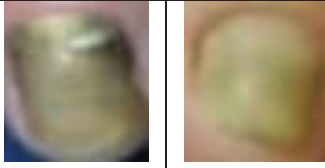
III. DATA COLLECTION

A vital step and among the most difficult tasks in deep learning is data collection. A robust dataset is essential for machine learning by neural networks practices in order to train the models. During testing, a model that has been properly taught performs more accurately.

For topics like nail diseases, which contain a wide range of photos that might improve the existing medical diagnosis standard, there are no open source datasets available. Because there isn't a dataset specifically for nail syndromes,

we created one by taking into account colour features, which includes about 18025 images from various disease categories (as presented in Table I).

TABLE I. DATA COLLECTION SAMPLES

Class name	Data Sample		Symptoms of Disease
Black colour nails			Vitamin B12 deficiency, impairment of kidney function, cancer.
Bluish colour nails			Hepatitis, liver disease, chronic obstructive pulmonary disease, atherosclerosis, cardiovascular disorders, thick blood.
Copper colour nails			Infections by fungi, arsenic poisoning.
Grey colour nails			Arthritis, malnutrition, postoperative effects, glaucoma, lung problem, enema.
Green colour nails			Bacillus infection, dermatitis.
Purple colour nails			Congenital problems, circulatory problems, oxygen deprivation.
Red colour nails			Brain haemorrhage, high blood pressure, heart diseases, lung disease.
White colour nails			Cardiovascular or lung complications, Hemorrhage, anaemic condition, kidney disease, hepatic impairment, helminth infection infestation.
Yellow colour nails			Liver disease, Diabetes, digestive problems.

We have chosen to use high-quality, appropriately coloured, and sized photos that correctly portray the various nail ailments. Additionally, we made use of the pre-trained model, AlexNet, which was developed using the ImageNet dataset's zillions of sample data. By putting the AlexNet architecture to use, we adjusted our dataset.

TABLE II. NUMBER OF IMAGES IN THE COLLECTION THAT RELATE TO VARIOUS NAIL CLASSES

Class name	Number of Images
Black colour nails	2612
Bluish colour nails	1449
Copper colour nails	3234
Grey colour nails	543
Green colour nails	1154
Purple colour nails	356
Red colour nails	1395
White colour nails	3778
Yellow colour nails	3504
Total	18025

The dataset's total number of data set that is associated with various nail syndromes is shown in Table II.

The datasets were divided into different training and testing sets to increase their efficacy. The training set was created by randomly choosing 80% of the sample data from each dataset's categories, with the remaining 20% going into the examination set.

IV. METHODOLOGY

We used a variety of Convolutional Neural Network Architecture to collect the necessary characteristics for classifying our nail disease class. On a computer with an Intel i5 Processor, 8GB GPU and 8GB RAM, our suggested model was run.

A. Features of Nails for Diseases Prediction

There are several features of nails that can be used for disease classification, including:

- **Nail shape:** The shape of the nail can provide diagnostic information about certain diseases. For example, clubbing of the nails is often seen in lung and heart disease.
- **Nail color:** Changes in nail color can indicate various diseases, such as yellowing of the nails in fungal infections or bluish discoloration in cyanosis.
- **Nail texture:** Changes in the texture of the nails can also provide diagnostic information. For example, thickened nails may indicate a fungal infection, while pitting or ridging may be a sign of psoriasis.
- **Nail plate thickness:** The thickness of the nail plate can be measured using ultrasound or other imaging

techniques and can provide information about nail disease.

- **Presence of nail lesions:** The presence of nail lesions such as onycholysis (separation of the nail plate from the nail bed), subungual hematoma (bleeding under the nail plate), or nail bed hyperkeratosis (thickening of the skin under the nail) can indicate various nail diseases.
- **Microscopic changes:** Microscopic examination of nail clippings or biopsy samples can reveal the presence of fungal or bacterial infections, or abnormalities in the nail matrix.

In addition to these features, machine learning algorithms can also be trained on other features extracted from nail images, such as texture features, shape features, and edge features, to classify nails into different disease categories. The combination of these features can provide a more comprehensive and accurate assessment of nail health and can help diagnose various nail diseases.

B. Nail Image Processing

Nail image processing for disease classification in humans is an important application of computer vision and machine learning techniques. The nails can provide valuable diagnostic information about various diseases, including fungal infections, psoriasis, and melanoma.

The process of nail image processing for disease classification typically involves the following steps:

- **Image acquisition:** Images of the nails are captured using a digital camera or a smartphone camera. The images should be of high quality and should capture the entire nail plate.
- **Preprocessing:** The images are preprocessed to enhance their quality and remove noise. This may involve techniques such as color normalization, contrast enhancement, and noise reduction.
- **Feature extraction:** Relevant features are extracted from the images to represent the nail structure and texture. This may involve techniques such as edge detection, texture analysis, and shape analysis.
- **Classification:** Machine learning algorithms are trained on the extracted features to classify the nail images into different disease categories. This may involve techniques such as support vector machines (SVMs), random forests, or convolutional neural networks (CNNs).
- **Validation:** The performance of the classification model is evaluated using metrics such as accuracy, precision, and recall. The model is then refined and re-evaluated until satisfactory performance is achieved.

It is important to note that nail image processing for disease classification is a complex task that requires expertise in both computer vision and dermatology. Additionally, the availability of high-quality annotated datasets is critical for developing accurate and reliable classification models. Therefore, collaboration between computer scientists and medical professionals is necessary to ensure the success of such applications.

C. Model Ensembling

Ensembling is a popular technique in deep learning for improving model performance by combining the outputs of multiple models. In the case of Convolutional Neural Networks (CNNs), ensembling can be particularly effective because CNNs are often used for image recognition tasks, where there may be a large amount of variation in the images. There are several approaches to ensembling CNN models:

- **Model averaging:** This involves training multiple CNN models on the same dataset, with different initializations or hyperparameters. The outputs of these models are then averaged to obtain a final prediction.
- **Bagging:** This involves training multiple CNN models on different subsets of the training data. The outputs of these models are then combined, typically by taking the average or the majority vote.
- **Boosting:** This involves training a sequence of CNN models, where each subsequent model is trained on the examples that the previous models misclassified. The final prediction is then obtained by combining the outputs of all the models.
- **Stacking:** This involves training multiple CNN models, where the output of each model is used as input to a meta-model that learns to combine the predictions of the base models.

Ensembling can be a powerful technique for improving the performance of CNN models, but it comes with some trade-offs. Ensembling requires more computational resources and can be more difficult to implement than training a single model. Additionally, ensembling may not always lead to improved performance, especially if the models being ensembled are too similar or if they are overfitting the training data. As with any technique in deep learning, ensembling should be evaluated on a case-by-case basis to determine whether it is appropriate for a given task.

Proposed methodology started by using our dataset to refine the pre-trained architecture: AlexNet. Additionally, proposed methodology evaluated the performance of collected data against a number of cutting-edge algorithms, including RF, KNN, and SVM, in order to establish a comparison between the accuracy attained by our suggested method and the current state-of-the-art algorithms.

D. Planned Methodology

With the methodology we propose, a set of 09 nail image classes are recognized and categorized using deep CNN. Using our precise dataset created specifically for nail illnesses, we implemented our own pre-trained architecture: AlexNet on the ImageNet dataset and fine-tuned it. Our model structural design has eight layers, including three fully connected layers and five convolutional layers.

The number of unseen layers in CNN typically ranges from 6 to 8. As indicated by [21], CNN boasts a whopping 60 million parameters, more than 600,000 neurons, and 600 million connections.

The equation below defines the conventional method of modeling a neuron:

$$f(p) = \tanh(p) \text{ or } f(p) = (1 + e^{-p})^{-1} \quad (1)$$

The process can be quite sluggish, which is why we implement RELU, also known as "Non-saturating nonlinearity," which is quicker in comparison. According to [16], a network of tanh neurons converges six times slower than a Convolutional Neural Network using RELU.

$$f(p) = \max(0, p) \quad (2)$$

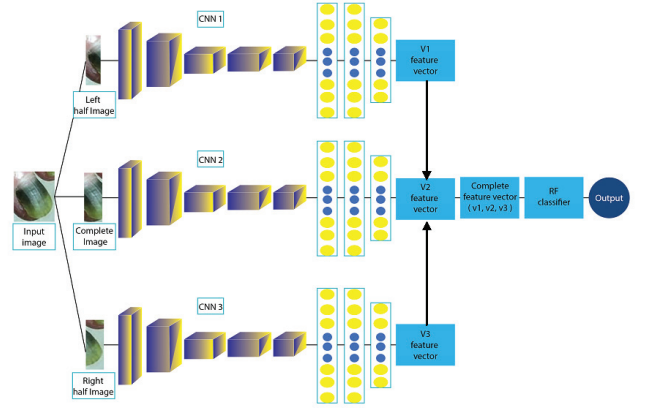


Fig. 1. Proposed Architecture

We used a hybrid strategy that makes use of three different CNNs in order to accurately capture the numerous features of the human nail. Each network's illustrations from the 7th entirely associated layer were taken out and saved as a 4096 by 1 vector. These three trajectories were then combined to create the concluding feature trajectory, designated as V, where V {V1, V2, V3} and the resulting feature vectors are V1, V2 and V3. The concluding feature trajectory, V, was subjected to the RF classifier, as shown in above figure. Utilizing cutting-edge algorithms like SVM, RF, KNN and CNN with SVM, we also assessed the precision of the fine-tuned dataset.

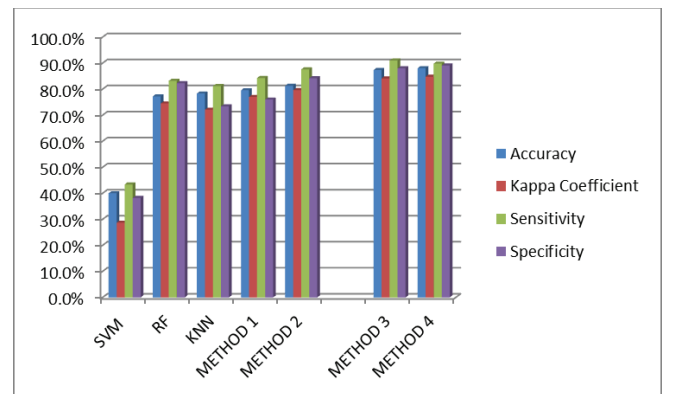


Fig. 2. Performance of Framework

By creating graphical representation, we assessed how well our framework performed [22]. This entails making a graph by graphing the "100-specificity" value and the sensitivity value on X-axis and Y-axis respectively. In proposed example, sensitivity was determined by how many pixels were correctly categorised as belonging to the "nail colour" class.

TABLE III. ACCURACY ASSESSMENT

Classification	Accuracy	Kappa Coefficient	Sensitivity	Specificity
SVM	40.01%	0.287	0.434	0.382
RF	77.22%	0.745	0.832	0.823
KNN	78.33%	0.721	0.812	0.734
METHOD 1	79.58%	0.77	0.843	0.76
METHOD 2	81.29%	0.796	0.876	0.842
METHOD 3	87.33%	0.841	0.91	0.88
METHOD 4	88.05%	0.848	0.898	0.891

V. RESULT

The numerical graphs in this part are intended to show the outcomes of the tests indicated above. For accuracy assessment, four alternative methodologies were taken into consideration. We found that method 3's accuracy was greatest (87.33%) and that our suggested approach's specificity and sensitivity values were both 0.88 and 0.91. With little pre-processing and refinement, the recital of the hybrid of Convolutional Neural Networks was shown to be equivalent to that of working dermatologists and cutting-edge machine learning techniques. Networks that were previously trained on further collected data containing relatively comparable data were used to generate the findings. To get these outcomes, we adjusted the Convolutional Neural Network architecture using data from our own dataset.

In technique 1, Convolutional Neural Network was applied on a whole picture of the nail. In technique number 2, we divided the image into its left and right halves and simultaneously fed each half to distinct two CNNs. One feature vector was created by combining the feature vectors collected from the two halves. This attribute trajectory was then used for Random Forest classification. In technique three, proposed method divided the image into two pieces once more, giving the left and right half to other Convolutional Neural Networks and the entire image to a separate Convolutional Neural Network.

The concluding organization was carried out using RF on a single feature vector created by combining the three feature vectors. In technique 4, we divided the image into four equal parts and gave each to a different CNN. Nevertheless, we found that the differences in accuracies between methods 3 (87.33%) and 4 (88.05%) were comparable during the correctness valuation. We recommend method 3 as the most effective strategy since state4 required 4 hours more for training than state 3.

We discovered that adding more convolutional layers to a structure improves accuracy because it allows for more accurate and effective extraction of feature.

VI. CONCLUSION

In order to categorize 09 nail picture classes for nail illnesses using our suggested strategy, we used an ensemble of Convolutional Neural Networks. We achieved an accuracy of 81.22%, which is comparable for existing techniques. The work was made more difficult by the absence of a pre-existing collected data with detailed pictures of nail disorders. But there is a chance for more expansion in subsequently.

Additionally, we want to develop a more comprehensive model have been used to find additional syndromes like hair

and dental conditions. Our research can be used to identify nail diseases early, which can result in prompt medical care. Dermatologists may expand and make use of our work to analyze patient data and identify disorders with less personnel and human participation.

REFERENCES

- [1] Wollina, Uwe, et al. "The Diagnosis and Treatment of Nail Disorders." *Deutsches Aertzteblatt International* 113 (2016)
- [2] Fletcher, C.L., Hay, R.J. and Smeeton, N.C., 2003. Observer agreement in recording the clinical signs of nail disease and the accuracy of a clinical diagnosis of fungal and non fungal nail disease. *British Journal of Dermatology*, 148(3), pp.558-562.
- [3] Langenbruch, A., Radtke, M.A., Krensel, M., Jacobi, A.Reich, K. and Augustin, M., 2014. Nail involvement as a predictor of concomitant psoriatic arthritis in patients with psoriasis. *British Journal of Dermatology*, 171(5), pp.1123-1128.
- [4] Sharma, V. and Ramaiya, M., 2015. Nail Color and Texture Analysis for Disease Detection. *International Journal of Bio-Science and Bio-Technology*, 7(5), pp.351-358.
- [5] Chen, R.H., Snorrason, M., Enger, S.M., Mostafa, E., Ko, J.M., Aoki, V. and Bowling, J., 2016. Validation of a skin-lesion image-matching algorithm based on computer vision technology. *Telemedicine and e-Health*, 22(1), pp.45-50.
- [6] Esteva, Andre, Brett Kuprel, and Sebastian Thrun. "Deep networks for early stage skin disease and skin cancerclassification." *Unpublishedmanuscript*(2015)
- [7] Cybenko, G., 1992. Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals, and Systems (MCCS)*, 5(4), pp.455-455.
- [8] Bengio, Yoshua, et al. "Towards biologically plausible deep learning." *arXiv preprint arXiv:1502.04156* (2015).
- [9] Sutskever, Ilya, Oriol Vinyals, and Quoc V. Le. "Sequence to sequence learning with neural networks." In *Advances in neural information processing systems*, pp. 3104-3112. 2014.
- [10] Deng, Jia, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. "Imagenet: A large-scale hierarchical image database." In *Computer Vision and Pattern Recognition, 2009. CVPR 2009. IEEE Conference on*, pp. 248-255. IEEE, 2009.
- [11] Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. "Deep learning. Book in preparation for MIT Press." URL< <http://www.deeplearningbook.org> (2016)
- [12] Krizhevsky, Alex, Ilya Sutskever, and Geoffrey E. Hinton. "Imagenet classification with deep convolutional neural networks." In *Advances in neural information processing systems*, pp. 1097-1105. 2012.
- [13] Indi, T.S. and Gunge, Y.A., 2016. Early Stage Disease Diagnosis System Using Human Nail Image Processing. *International Journal of Information Technology and Computer Science (IJITCS)*, 8(7), p.30.
- [14] Liao, H., 2016. A deep learning approach to universal skin disease classification. University of Rochester Department of Computer Science, CSC.
- [15] Paradowski, M., Markowska-Kaczmarska, U., Kwasnicka, H. and Borysewicz, K., 2009, September. Capillary abnormalities detection using vessel thickness and curvature analysis. In *International Conference on Knowledge-Based and Intelligent Information and Engineering Systems* (pp. 151-158). Springer, Berlin, Heidelberg.
- [16] Dhandra, B., et al. "Color Histogram Approach for Analysis of Psoriasis Skin Disease." *Int. Conf. on Multimedia Processing*. 2013.
- [17] Cortes, C. and Vapnik, V., 1995. Support vector machine. *Machine learning*, 20(3), pp.273-297.
- [18] Lee, Y., 1991. Handwritten digit recognition using k nearest-neighbor, radial-basis function, and backpropagation neural networks. *Neural computation*, 3(3), pp.440-449.
- [19] Bosch, A., Zisserman, A. and Munoz, X., 2007, October. Image classification using random forests and ferns. In *Computer Vision, 2007. ICCV 2007. IEEE 11th International Conference on* (pp. 1-8). IEEE.
- [20] He, K., Zhang, X., Ren, S. and Sun, J., 2015. Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In *Proceedings of the IEEE international conference on computer vision* (pp. 1026-1034).

- [21] http://vision.stanford.edu/teaching/cs231b_spring1415/slides/alexnet_tugce_kyunghee.pdf
- [22] Pontius Jr, R.G. and Millones, M., 2011. Death to Kappa: birth of quantity disagreement and allocation disagreement for accuracy assessment. *International Journal of Remote Sensing*, 32(15), pp.4407-4429.
- [23] Sirguey, P., Mathieu, R., Arnaud, Y., Khan, M.M. and Chanussot, J., 2008. Improving MODIS spatial resolution for snow mapping using wavelet fusion and ARSIS concept. *IEEE Geoscience and Remote Sensing Letters*, 5(1), pp.78-82.
- [24] Costa A, Molnar P, Stutenbecker L, Bakker M, Silva TA, Schlunegger F, Lane SN, Loizeau JL, Girardclos S. Temperature signal in suspended sediment export from an Alpine catchment. *Hydrology and Earth System Sciences Discussions*. 2017.